



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

SENIOR CERTIFICATE EXAMINATIONS/ NATIONAL SENIOR CERTIFICATE EXAMINATIONS

ENGINEERING GRAPHICS AND DESIGN P1

MAY/JUNE 2024

MARKING GUIDELINES

MARKS: 100

These marking guidelines consist of 8 pages.

ANSWERS		
1	CHECKED	1
2	1 : 500	1
3	08/05/2023	1
4	CONTRACTORS	1
5	6	1
6	2	1
7	INSPECTION EYE	1
8	<u>CLIFF</u> FACE	1
9	SHRUBBERY	1
10	BUILDING LINE	1
11	MUNICIPAL <u>WATER SUPPLY</u>	1
12	1,8	1
13	100	1
14	<u>DIRECTION</u> OF <u>FLOW</u>	1
15	WATER METER	1
16	10	2
17	GREEN	1
18	SSW / SOUTH SOUTH WEST / SOUTH WEST / SW	2
19	See below	3
20		3
21		4
TOTAL		30

ANSWER 19

Show ALL calculations.

APPLYING CORRECT FORMULA ✓

$$S = P - (S + S + S + S)$$

$$= 173,56 - (48,04 + 47,96 + 25,95 + 34,58)$$

$$= 17,03 \quad \checkmark \quad \checkmark$$

ANSWER 20

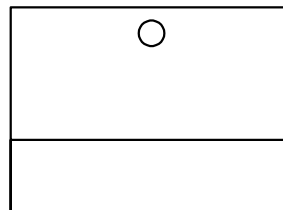
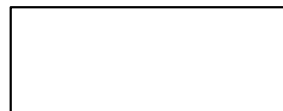
Show ALL calculations.

APPLYING CORRECT FORMULA ✓

$$A = (L \times B) + (L \times B) + \dots$$

$$= (30 \times 13) + (5 \times 10) + (21 \times 10,5) + (9 \times 2)$$

$$= 678,5 \text{ m}^2 \quad \checkmark \quad \checkmark$$

ANSWER 21

FREEHAND

1

RECTANGLE

✓ 1

TOP
RECTANGLE

✓ 1

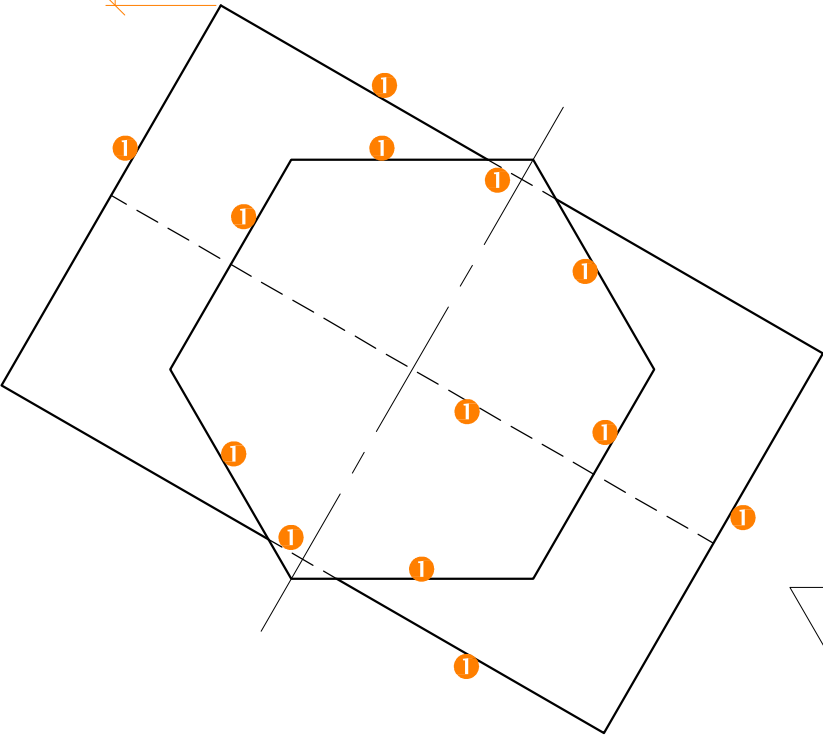
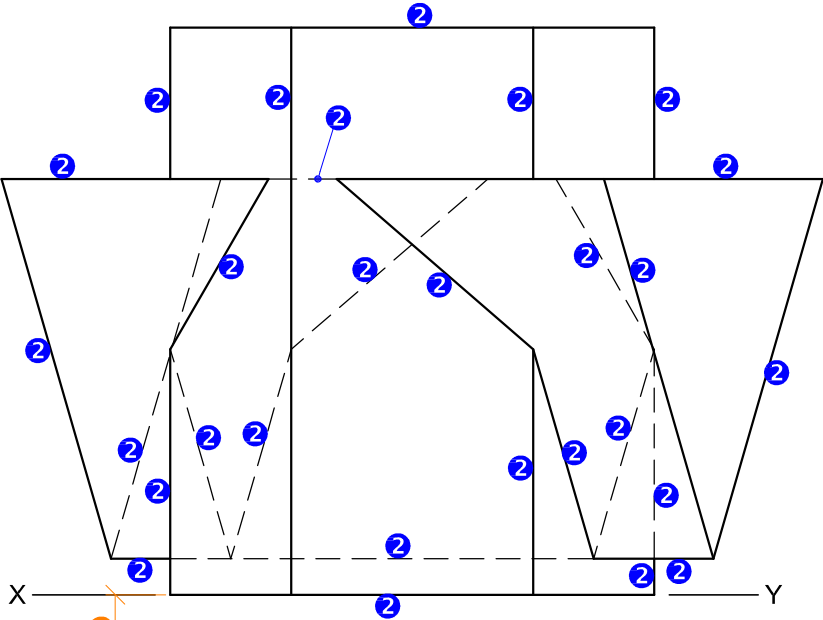
BOTTOM
RECTANGLE

✓ 1

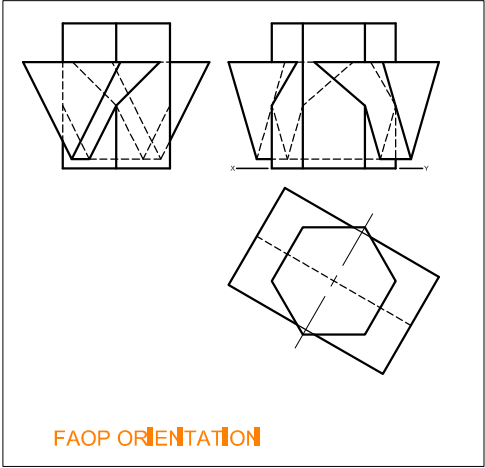
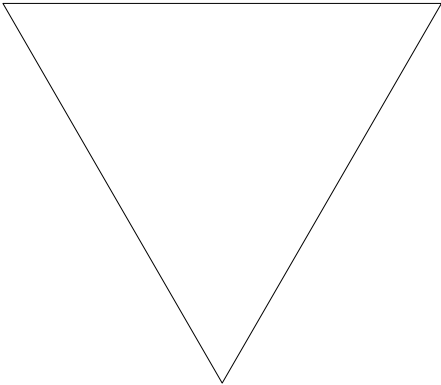
CIRCLE

1

PAPER 1 QUESTION 1
GRADE 12
DBE MAY/JUNE 2024
MARKING GUIDELINES



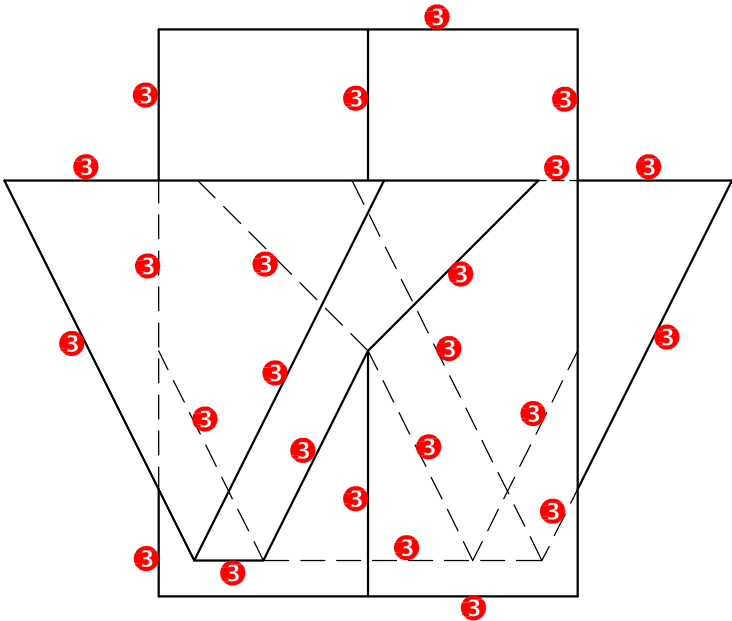
1. CONSTRUCTION
- 30° ANGLE1
- SHAPE OF HEXAGON1
- SHAPE OF TRIANGLE1
- TOTAL3



FAOP ORIENTATION

ASSESSMENT CRITERIA		
1	TOP VIEW	10
2	FRONT VIEW	14
TOTAL		24

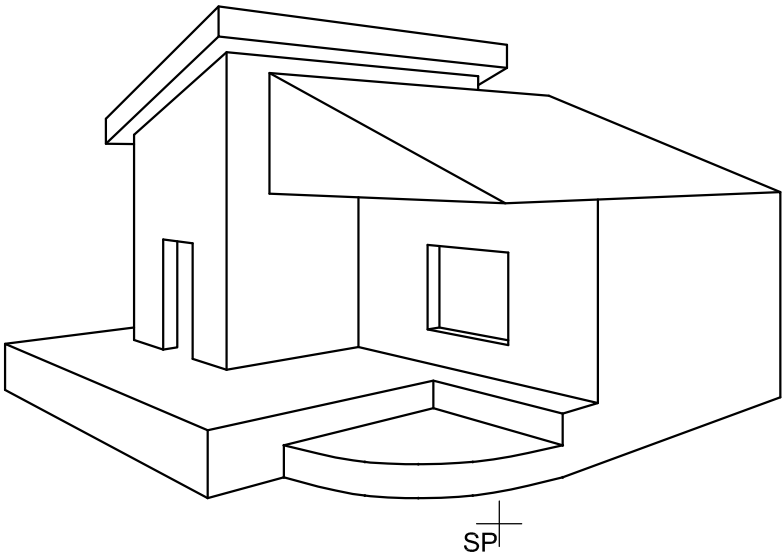
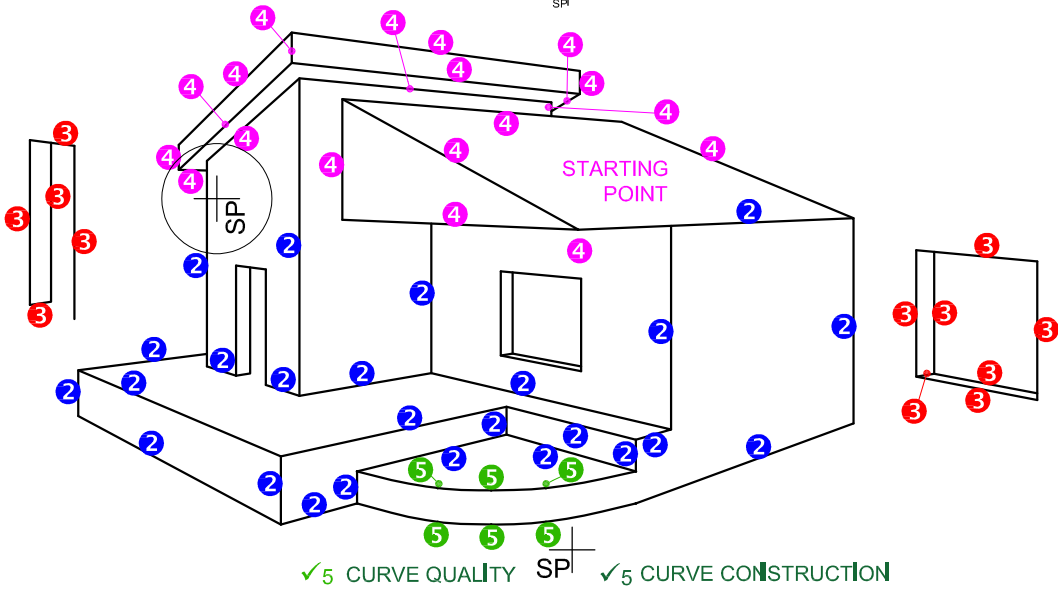
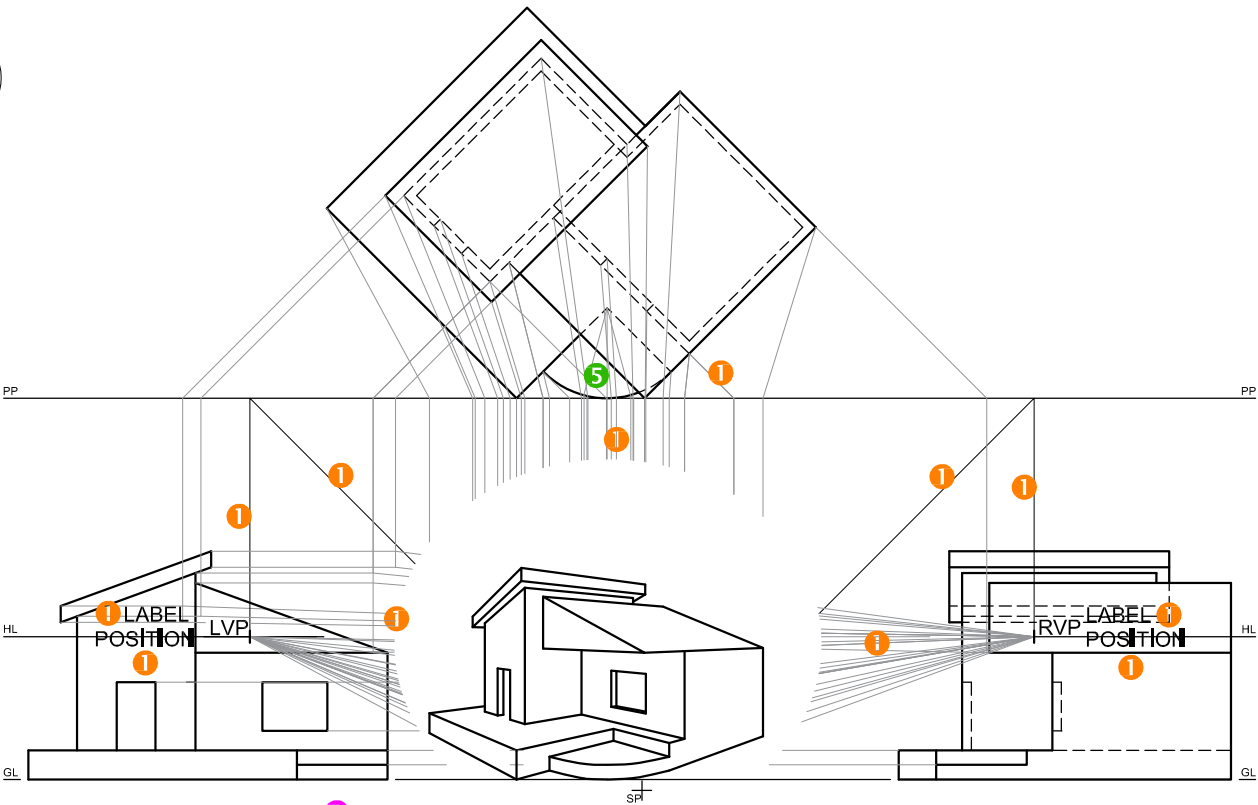
PAPER 1 QUESTION 2
GRADE 12
DBE MAY/JUNE 2024
MARKING GUIDELINES



ASSESSMENT CRITERIA		
3	RIGHT VIEW	12
TOTAL		12

PAPER 1 QUESTION 2
GRADE 12
DBE MAY/JUNE 2024
MARKING GUIDELINES

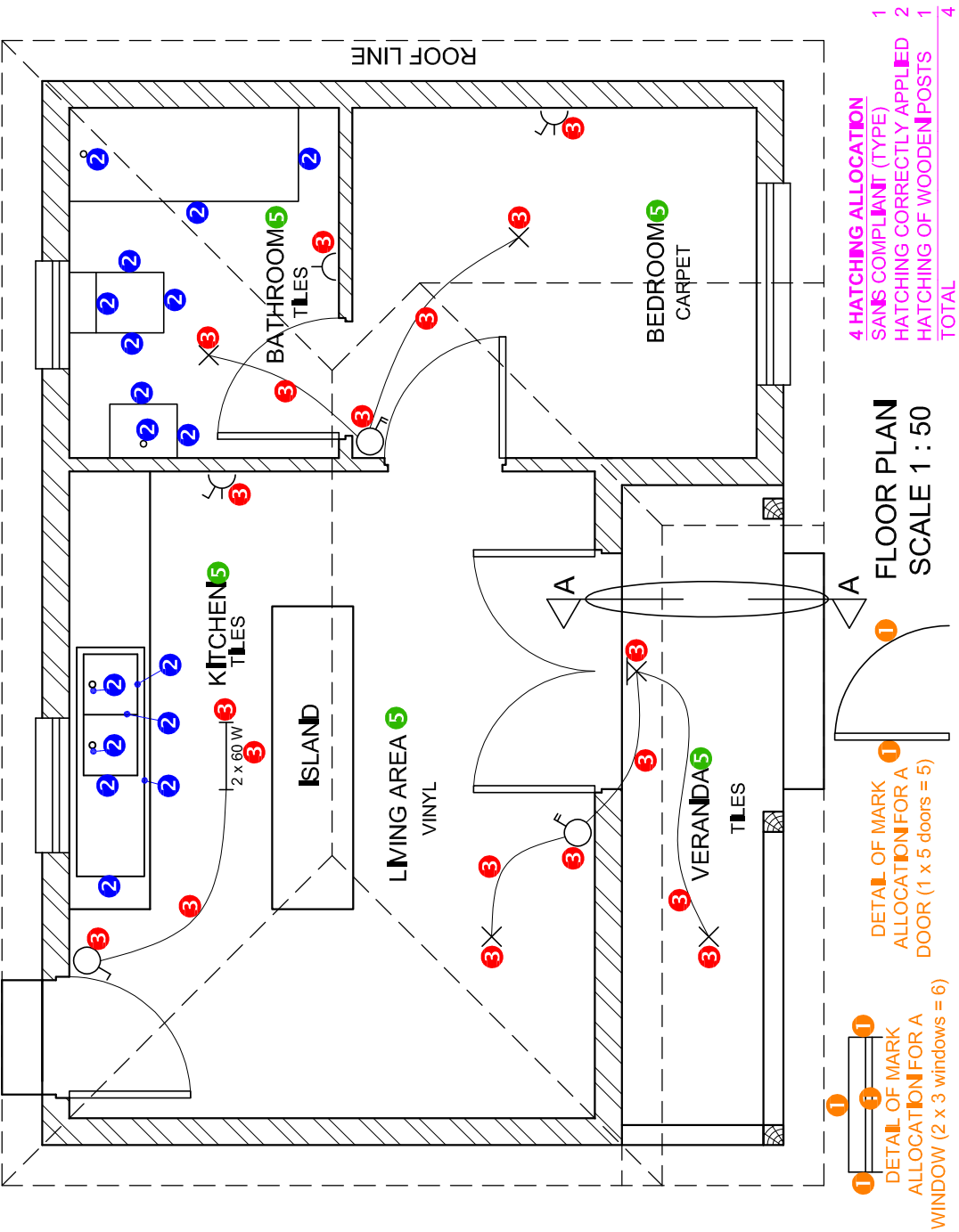
RVP



ASSESSMENT CRITERIA		
1	CONSTRUCTION	6
2	WALL + BASE + STAIRS	12 ¹ / ₂
3	WINDOW + DOOR	6
4	ROOF	9
5	ARC	5 ¹ / ₂
TOTAL		39

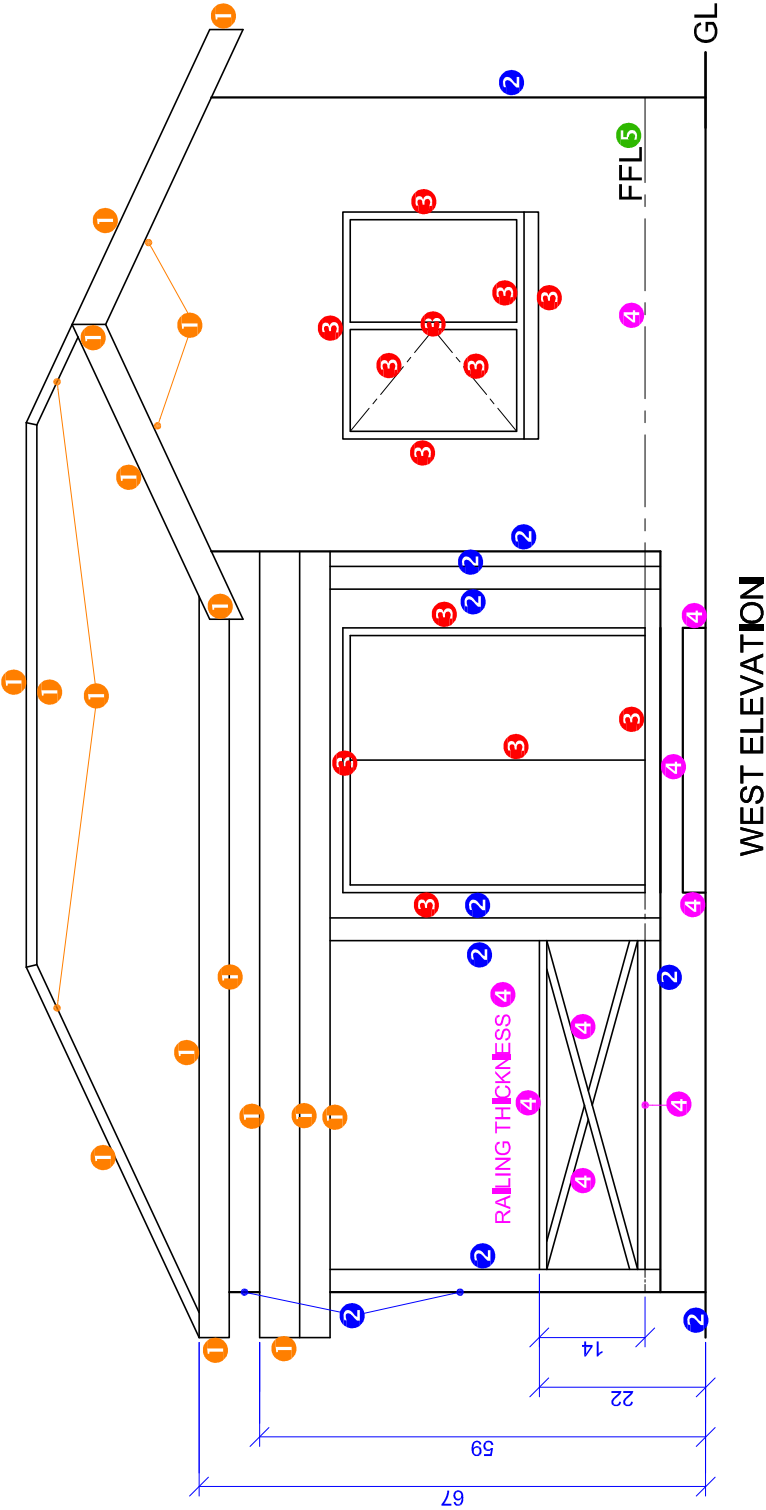
PAPER 1 QUESTION 3
GRADE 12
DBE MAY/JUNE 2024
MARKING GUIDELINES

LVP



ASSESSMENT CRITERIA	
FLOOR PLAN	
1	DOORS + WINDOWS 11
2	FEATURES 8 1/2
3	ELECTRICAL 9 1/2
4	HATCHING 4
5	LABELS 2 1/2
SUBTOTAL 35 1/2	

PAPER 1 QUESTION 4
GRADE 12
DBE MAY/JUNE 2024
MARKING GUIDELINES



ASSESSMENT CRITERIA		
WEST ELEVATION		
1	ROOF + LEANING WALLS + POSTS + VERANDA	8 ¹ / ₂
2	DOORS + WINDOW	5
3	STEP + FFL + RAILING	6 ¹ / ₂
4	LABELS	4 ¹ / ₂
5		1 ¹ / ₂
SUBTOTAL		25

PAPER 1 QUESTION 4
GRADE 12
DBE MAY/JUNE 2024
MARKING GUIDELINES

1. MARK ALLOCATION FOR ROOF

A: ROOF ANGLE	1
B: ROOF SHEET	1
C: PURLIN	1
D: TRUSS + OVERHANG	3
E: WALL PLATE	1
F: BRANDING	0.5
G: CEILING BOARD	0.5
H: FASCIA	1.5
J: LEAN TO ROOF	5.5
TOTAL	15

3. WOOD HATCHING ALLOCATION

SANS COMPLIANT (TYPE)	0.5
HATCHING CORRECTLY APPLIED	0.5
TOTAL	1

The drawing shows a detailed section of a roof and wall. The roof is supported by a truss system (D) and has an overhang. The wall is constructed with a wall plate (E), branding (F), ceiling board (G), and fascia (H). The roof is covered with roof sheets (B) and has a lean-to roof (J). The wall has a railing (4) and a DPC (5). The drawing includes dimensions such as 55, 35, and 4. Labels A through J are used to identify specific components. Hatching is used to represent different materials like concrete (H3) and brickwork (H4). Dimensions are given in millimeters.

ASSESSMENT CRITERIA		
DETAILED SECTION		
1	ROOF DETAIL	15
2	SLAB + WALL + STEPS + B/L	10
3	HATCHING	5 $\frac{1}{2}$
4	RAILING	3
5	LABELS	1
SUBTOTAL		34 $\frac{1}{2}$

PAPER 1 QUESTION 4
GRADE 12
DBE MAY/JUNE 2024
MARKING GUIDELINES